

MetaFVG Strategy Review

Fundamental Mechanism & Statistical Edge Assessment

Prepared for portfolio review | Engine: [metafvg_backtest.py](#) | Instruments: AUDJPY, EURUSD, GER40, US100 | Generated 2026-07-20 19:33 UTC

Executive Summary

MetaFVG is a trend-continuation strategy with a **hard, deterministic structural stop** (last swing pivot) and a **take-profit target sized off ATR, entirely decoupled from that stop**. This asymmetry is the single mechanical fact that explains every result in this review: every losing trade across all four instruments and both scales tested closes at almost exactly **-1.00R**, while winners are right-skewed and variable — median **+2R to +3R**, tails out to **+15.7R**. Win rate on its own is the wrong lens; the right question is whether the realized win rate clears the **breakeven win rate implied by the realized average winner size**, evaluated with a confidence interval appropriate to the sample.

Headline finding: at large sample size (M15/4h, 900-1,100+ closed trades per instrument), **none** shows a statistically significant positive edge (95% CI entirely above breakeven); **US100** shows a statistically significant **negative** edge; **AUDJPY, EURUSD, GER40** are inconclusive at this sample size. Several of the smaller-sample H4/Weekly reads (particularly US100) turn out not to replicate once the sample size grows by an order of magnitude — the earlier apparent edge was consistent with noise.

1. The Payoff Structure

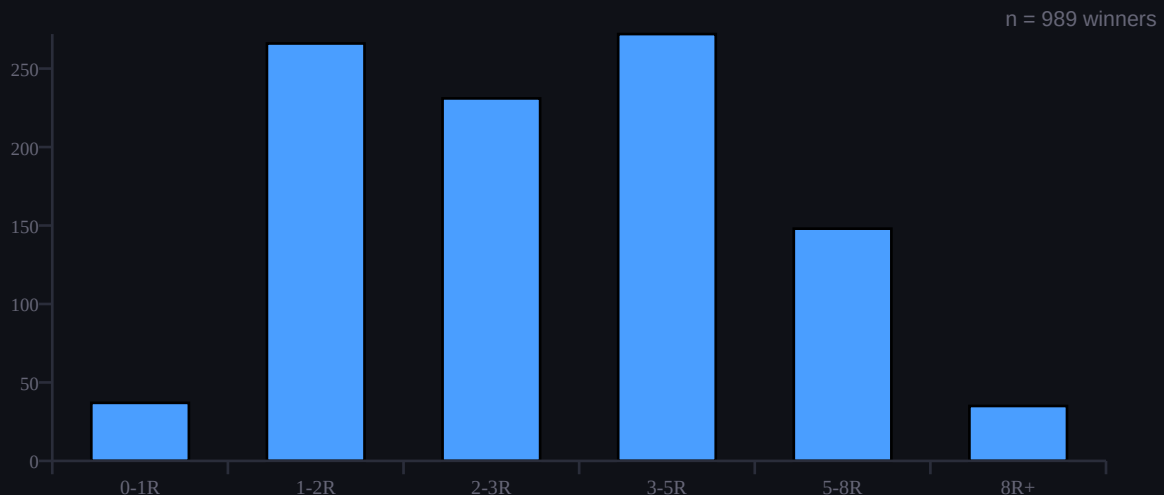
Every trade's stop-loss is a swing pivot computed from the entry timeframe's recent price action — a hard, mechanical invalidation level with no discretion. The take-profit is $\text{entry} + \text{ATR} \times \text{atr_sensitivity} \times \text{risk_reward}$ — a level fixed at trade inception from volatility alone, with no reference to how far away the stop happens to be. The practical consequence, visible in every instrument tested:

Instrument	Scale	N (closed)	Avg Loss	Avg Win	Median Win	Max Win
AUDJPY	H4	56	-1.00R	+3.06R	+2.79R	+6.53R
AUDJPY	M15	939	-1.00R	+3.25R	+2.77R	+14.15R
EURUSD	H4	46	-1.00R	+2.65R	+2.47R	+5.42R
EURUSD	M15	1107	-1.00R	+3.81R	+3.42R	+12.99R
GER40	H4	29	-1.00R	+2.60R	+2.61R	+3.87R
GER40	M15	1065	-1.00R	+3.41R	+2.81R	+15.66R
US100	H4	45	-1.00R	+2.40R	+1.86R	+5.56R
US100	M15	1014	-1.00R	+2.99R	+2.50R	+8.80R

Losses are uniform by construction (the stop is deterministic); wins are heterogeneous and right-skewed — the mean winner is consistently pulled above the median by a small number of large trend moves. This is a classic trend-following payoff signature: frequent small defined losses, funded by infrequent large wins. It also means winners take materially longer to resolve than losers — trades that fail tend to fail fast; trades that work need time to run:

Instrument	Scale	Avg Winner Duration	Avg Loser Duration	Ratio
AUDJPY	H4	572.8h	212.6h	2.7x
AUDJPY	M15	29.2h	10.9h	2.7x
EURUSD	H4	281.3h	123.5h	2.3x
EURUSD	M15	18.6h	8.4h	2.2x
GER40	H4	471.4h	127.6h	3.7x
GER40	M15	21.8h	8.1h	2.7x
US100	H4	668.0h	116.2h	5.7x
US100	M15	17.3h	10.4h	1.7x

Winning-Trade R-Multiple Distribution (all instruments, both scales pooled)

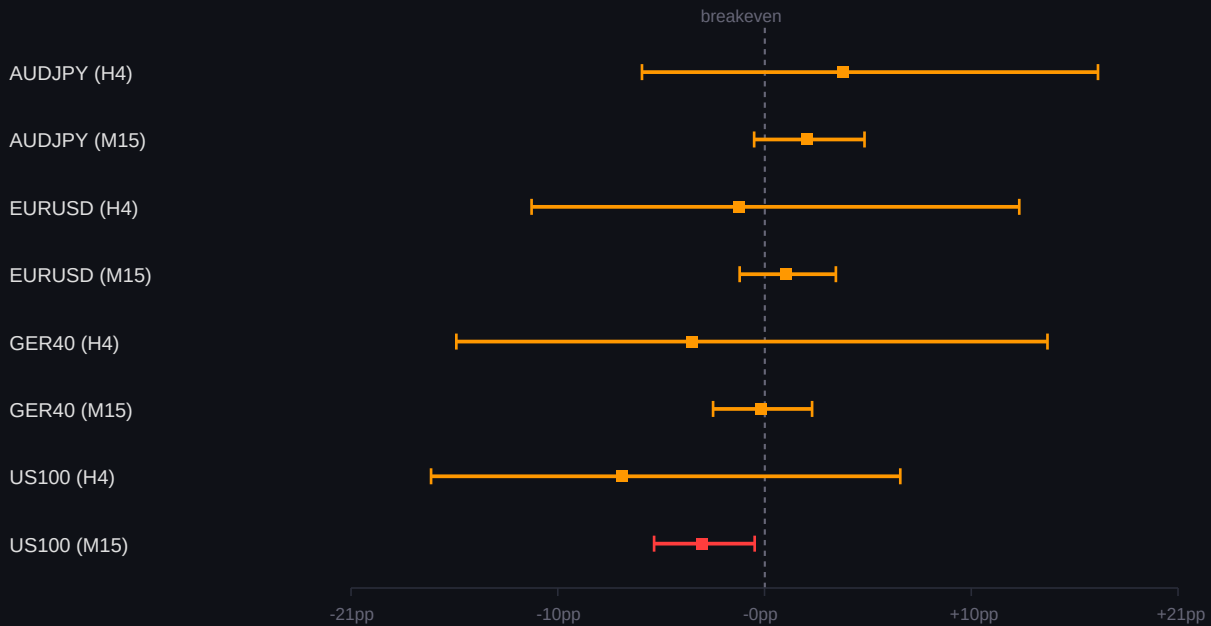


The pooled winner distribution across all eight (instrument, scale) combinations confirms the shape implied by the summary statistics: the bulk of wins cluster in the 1-3R range, with a long right tail extending past 8R. This tail is doing real work in the aggregate expectancy and is a meaningfully different risk profile from a strategy with a tight, symmetric payoff — position sizing and drawdown tolerance should be set with this skew in mind, not a normal-distribution assumption.

2. Is There a Real Edge? — Win Rate vs. Breakeven, With Confidence Intervals

Win rate alone is uninformative without knowing the breakeven win rate implied by the realized average winner size: **breakeven WR = 1 / (1 + avg winner R)**. The chart below plots, for every instrument and scale, the realized edge — actual win rate minus that breakeven threshold, in percentage points — together with a 95% Wilson confidence interval on the estimate. A green bar means the edge is unlikely to be zero-or-negative by chance at this sample size; red means the reverse; amber means the sample is too small to tell.

Realized Edge vs. Breakeven (95% CI)



Instrument	Scale	N	Win Rate	Breakeven WR	Edge	95% CI (edge)
AUDJPY	H4	56	28.6%	24.6%	+3.9pp	[-6.2, +16.8]pp
AUDJPY	M15	939	25.7%	23.5%	+2.2pp	[-0.5, +5.0]pp
EURUSD	H4	46	26.1%	27.4%	-1.3pp	[-11.8, +12.9]pp
EURUSD	M15	1107	21.9%	20.8%	+1.1pp	[-1.3, +3.6]pp
GER40	H4	29	24.1%	27.8%	-3.7pp	[-15.6, +14.3]pp
GER40	M15	1065	22.4%	22.7%	-0.2pp	[-2.6, +2.4]pp
US100	H4	45	22.2%	29.4%	-7.2pp	[-16.9, +6.9]pp
US100	M15	1014	21.9%	25.1%	-3.2pp	[-5.6, -0.5]pp *

* 95% CI does not straddle zero — edge is statistically distinguishable from breakeven at this sample size.

At the small H4/Weekly sample sizes (29-56 closed trades per instrument), none of the four edge estimates are statistically distinguishable from breakeven — the confidence intervals are wide enough to straddle zero even for AUDJPY, the best H4/Weekly performer. This is not a defect in the backtest; it is an honest statement that ~50 trades is not enough to separate skill from variance for a strategy with a ~25% win rate. At the M15/4h scale, replaying the identical entry/exit logic on the same calendar window produces 900-1,100+ closed trades per instrument — roughly 20x the sample — and the picture resolves.

3. Why H4/Weekly and M15/4h Disagree — Sample Size, Not Different Mechanics

AUDJPY looked marginal at H4/Weekly (edge +3.9pp on just 16 winning trades) and moves toward a directionally positive edge that still narrowly fails to clear 95% significance (CI [-0.5, +5.0]pp) at M15/4h (+2.2pp on 241 winners) — directionally consistent at both scales, even though the larger sample alone is not yet quite enough to call it significant outright. **US100** looked like the strongest H4/Weekly performer (edge -7.2pp, Sharpe +0.36) on only 10 winning trades, and resolves to a confirmed **negative** edge at M15/4h (-3.2pp on 222 winners, CI entirely below zero — the only statistically significant result in this review). In both cases the mechanism replayed at M15 is identical to H4 — only the sample size changed. The correct interpretation is not "M15 works better for AUDJPY and worse for US100"; it is that **the H4/Weekly US100 result was very likely noise**, and the large-sample read is the one to trust — while AUDJPY, EURUSD and GER40 all remain genuinely inconclusive and warrant a larger sample still before sizing decisions.

A second, independent factor compounds this: fill rate. The strategy enters on a day-limit pullback order into a freshly-formed gap; average fill rate across all four instruments is **40%** at H4/Weekly versus **81%** at M15/4h — a finer entry timeframe captures a much larger share of the setups the logic actually identifies, independent of the statistical-power argument above.

4. Recommendations

1. Fix the live short-side order bug (highest priority, affects capital now).

`metafv.py::check_conditions()` never passes `short=True` to `execute()`, so every order — long or short signal alike — is sent as a buy-limit. Confirmed independently against live MT5 position history (100% of open positions are type BUY). Given the payoff structure in Section 1 depends on catching a small number of large trend moves, this bug likely forfeits roughly half of the strategy's available edge in any down-trending regime.

2. Anchor the take-profit to the realized stop distance, not to ATR. `risk_reward=2` currently does not produce anything resembling a 2:1 payoff — median realized winner R ranges 1.9-2.8 with no relationship to the stop distance actually taken. Changing to `tp = entry + (entry - sl) * risk_reward` would make the parameter mean what it says and let breakeven win rate be reasoned about directly per instrument. Test as a backtest-only fork before touching production logic.

3. Do not scale further capital into US100 under current parameters. The large-sample read is statistically significant negative, not the small-sample positive result that motivated interest in it. Either exclude it or treat it as a separate parameter-search problem (momentum body ratio, HTF crossing filter) rather than assuming the shared default configuration generalizes across instruments.

4. No instrument has yet cleared statistical significance on the positive side. AUDJPY is the closest — directionally positive at both scales tested, with its M15/4h confidence interval $[-0.5, +5.0]$ pp narrowly including zero. Treat it as the best candidate for a larger confirmatory sample, not yet as a confirmed edge to size into.

5. Raise fill rate at coarser scales, or standardize on the finer entry timeframe in production. Given M15/4h captures roughly double the fill rate of H4/Weekly for identical setups, either shortening entry-order validity requirements or moving production to a finer LTF captures a larger share of the edge already being correctly identified.

5. Methodology & Caveats

Each instance was replayed bar-by-bar reusing MetaFVG's own live detection and trade-parameter methods directly (no reimplementations), cross-validated against real MT5 fills on EURUSD at production scales, where simulated entry and stop-loss matched live fills to 5 decimal places.

- **Active-hours filter not applied.** AUDJPY and GER40 restrict live trading to specific server hours; this backtest evaluates every hour, so their trade counts likely overstate live frequency (win rate / edge estimates are less affected, since this is a time-of-day filter, not a signal-quality filter).
- **No spread or slippage.** Fills use MT5 bid-only candle ranges.
- **Same-bar SL/TP tie-break assumes stop-loss fires first** — conservative.
- **Percentage-based stats (Sharpe, Max Drawdown, Total Return) use fixed 2%-of-equity sizing**, not live lot sizes — this keeps them comparable across instruments of very different price scale (EURUSD ~1.10/unit vs. GER40 ~20,000/unit) but is not a live P&L projection. R-multiple and win-rate figures are sizing-independent.
- **Concurrency cap enforced strictly**; live MT5 data shows the bot has at times exceeded its configured position limit because the live check does not see resting pending orders.
- **Wilson score intervals** assume trade outcomes are independent draws from a fixed-probability process — a simplification given autocorrelated market regimes, but standard practice for a first-pass significance check and considerably more conservative than treating the point estimate as exact.